

VVU IVE

Immersive Virtual Environment

Evidence-Verification Layer for
Infrastructure Observations

SIMULATION DATA — NOT MUNICIPAL OPERATIONAL DATA

Executive Documentation Bundle

Prepared for: Department of Water and Sanitation (DWS)

From: Venture Vision Ubuntu (VVU)

Date: 2026-08-26

Repository: github.com/divhanimajokweni-ctrl/proofbridge-liner

Branch: feat/vvu-gov-deploy

Domain: vvu-gov.space-z.ai

Table of Contents

#	Folder	Document	Page
01	Executive	01a One-Page Executive Brief	4
01	Executive	01b Technical Demonstration Brief	5
02	Scientific	02a HOM — Hydraulic Observability Model	7
02	Scientific	02b Sparse Sensor Hypothesis	8
02	Scientific	02c EIS v1.0 — Evidence Independence Spec	9
02	Scientific	02d Architecture Figure	11
03	Software Evidence	03a VRES v1.0 Component Inventory	12
03	Software Evidence	03b Repository Verification	13
04	Water Demo	04a Leakage Validation Brief	14
04	Water Demo	04b NMBM Sandbox Specification	15
04	Water Demo	04c Hydraulic Incident Replay Script	16
05	Pilot	05a 72-Hour Validation Protocol	17
05	Pilot	05b Data Requirements	19

01a · One-Page Executive Brief

From: Venture Vision Ubuntu (VVU) · To: DWS Technical Validation · Date: 2026-08-26

What VVU IVE Is

VVU IVE is an **evidence-verification layer** for infrastructure observations. It allows sparse municipal observations to be correlated, independently assessed, provenance-tracked, and converted into auditable engineering evidence.

Why Water-Loss Matters

Municipal water networks lose significant volumes to non-revenue water (NRW) — underground leaks that are difficult to locate with sparse sensor coverage. SCADA systems generate measurements, but the engineering challenge is converting those measurements into verified, auditable evidence that supports maintenance decisions.

What We Have

- **Working software** — zero lint errors, zero console errors, healthy routes, reproducible build
- **EIS v1.0 engine** — Evidence Independence Scoring that prevents evidence inflation (5 correlated sensors $\neq$ 5 independent proofs)
- **Hydro-Bayesian Kernel (HBK)** — Sequential Bayesian localization over 32×32 grid, MAP estimate + 95% credible radius
- **Sandbox pipeline** — setup.sh + run.sh generates auditable evidence files
- **Provenance chain** — every observation carries 11 fields: sensor ID, firmware, calibration, timestamp, location, DMA, environment, processing, attestation, quality, type

What We Have NOT Done

VVU has **not** operated a municipal water distribution network. We do not represent the current prototype as having been validated against municipal operational data. All water demonstration data is clearly labelled as **SIMULATION / PLACEHOLDER**.

What We Seek from DWS

Domain validation. Does the engineering model VVU has encoded map correctly to how DWS and municipalities actually observe, diagnose, and verify infrastructure problems?

If DWS provides governed historical or test data, VVU will run the system against it under a 72-hour validation protocol and measure whether it identifies the patterns DWS engineers already recognise.

"I haven't operated a municipal water network myself. My practical experience is on the systems-engineering and software side. I built IVE to address an evidence and verification problem, and I'm here to test whether the assumptions we've encoded correspond to how DWS and municipalities actually observe, diagnose, and verify infrastructure problems."

01b · Technical Demonstration Brief

Audience: DWS technical engineers and validators · Classification: SIMULATION DATA

What VVU IVE Is

VVU IVE (Immersive Virtual Environment) is an **evidence-verification layer** for infrastructure observations. It sits around existing SCADA and operational systems — it does not replace them.

The Engineering Chain

Municipal water network

- sparse observations (flow, pressure, level, pump/valve status)
- anomaly detection (hydraulic deviation from baseline)
- evidence correlation (link related observations across sensors + time)
- independence assessment (are corroborating observations genuinely independent?)
- candidate location inference (narrow the search area)
- field verification (human confirms or rejects)
- auditable evidence record (complete provenance chain)

EIS v1.0 — Evidence Independence Scoring

The system does **NOT** count "5 sensors agree = 100% confidence." It asks: are those 5 observations genuinely independent, or are they correlated measurements of the same event?

Scoring: PRIMARY (0.3) + CORRELATED (0.2) + INDEPENDENT (0.4) = 0.9 confidence → VERIFIED_CANDIDATE

5 pressure sensors on the same DMA = 1 correlated hydraulic event. 1 flow anomaly + 1 field observation + 1 acoustic signal = 3 independent evidence types.

HBK — Hydro-Bayesian Kernel

Sequential Bayesian localization over a 32×32 candidate-location grid. $P(S_t | O_{1:t}) \propto P(O_t | S_t) \cdot P(S_t | S_{t-1})$. Location is inferred from distance-attenuated acoustic/pressure amplitude at fixed listening taps. Mining-blast noise is handled as a Poisson mixture and down-weighted (not excluded) to suppress false positives.

Zero Fabrication Rule

Missing data is never guessed. It is flagged as UNDEFINED. The system may not manufacture evidence. The 11-field provenance spine per observation ensures every conclusion traces to specific, auditable observations.

Repository State (Verified)

Check	Result
Lint	✓ 0 errors, 0 warnings
Console errors	✓ 0

Dev server	✓ All routes 200
Reproducible build	✓ git clone → bun install → bun run dev
Sandbox pipeline	✓ setup.sh + run.sh → /evidence
Custom domain	vvu-gov.space-z.ai

Proposed DWS Validation

A **72-hour validation protocol** where VVU IVE is run against a DWS-provided historical or test dataset from a single DMA. Success criteria: detected events, lead time, false alerts, missed events, independence scores.

02c · EIS v1.0 — Evidence Independence Specification

Purpose

EIS v1.0 defines how VVU IVE evaluates whether observations that appear to corroborate each other are **genuinely independent** — or are merely correlated measurements of the same underlying phenomenon.

Evidence States

State	Meaning	Example
Valid	Within physical range, quality VALID	Flow 111 L/s with VALID flag
Missing	No data for 17+ minutes	Sensor offline & UNDEFINED
Anomalous	Impossible physics	999m pressure spike
Correlated	Same DMA, time, type	2 pressure sensors on same DMA
Independent	Different principle, location	Flow + field + acoustic
Insufficient	Not enough to assess	Only 1 sensor, no field evidence

Independence Scoring

```
Score = (has_primary × 0.3) + (has_correlated × 0.2) + (has_independent × 0.4)
```

```
If has_primary AND has_pump_context → REJECTED (false positive)
```

```
If score ≥ 0.8 → VERIFIED_CANDIDATE
```

```
Else → INSUFFICIENT_EVIDENCE
```

Key Rule: Correlated observations cap out

Multiple correlated sensors add **small** confidence (proves event happened), but they do NOT scale linearly. 1 flow anomaly + 5 correlated pressure drops ≠ 6 independent proofs. It equals 1 primary + 1 correlated event.

Key Rule: Independent corroboration is the gold standard

1 flow anomaly + 1 acoustic signal = **higher confidence** than 1 flow anomaly + 10 pressure sensors on the same DMA. Different measurement principles provide genuinely new information.

DMA Calibration

Parameter	Default	Range	Purpose
flowDeviationThreshold	10%	1-50%	Min flow deviation to flag
pressureDropThreshold	5%	1-30%	Min pressure drop to correlate
correlationTimeWindow	60 min	1-1440	Max time gap for correlation

Parameters are serialized into the audit receipt so the result is mathematically reproducible — a municipal investigator can verify exactly why the system made its decision.

Zero Fabrication Rule

Missing data is never guessed. It is flagged as UNDEFINED. If a sensor is offline, EIS does not interpolate a value and treat it as evidence. The observation count drops, and the independence score is recomputed from

remaining observations.

The system may not manufacture evidence.

04a · Leakage Validation Brief

The Chain

Municipal water network → sparse observations → anomaly detection → evidence correlation → independent evidence assessment → location/claim → verification → maintenance decision → auditable evidence record

Hydraulic Incident Replay

The 10-step interactive replay demonstrates the full evidence pipeline: baseline → anomaly injection → field evidence → acoustic evidence → operational context → correlation → independence assessment → claim → verification → audit trail.

Step	Action	Result
1	BASELINE	97 L/s @ 48.5m — nominal
2	INTRODUCE ANOMALY	Flow ±; 111 L/s, Pressure ±; 46.1m
3	ADD FIELD EVIDENCE	Ground moisture at segment S-142
4	ADD ACOUSTIC	Anomalous signal at S-142
5	CHECK CONTEXT	No pump/valve changes
6	CORRELATE	5 observations linked
7	ASSESS INDEPENDENCE	Score 1.00 ±; VERIFIED_CANDIDATE
8	GENERATE CLAIM	Zone S-142, DMA-7
9	VERIFY	Field confirms leak at pipe joint
10	AUDIT	SHA-256 receipt + 11-field provenance

05a · 72-Hour Validation Protocol

Objective

Validate whether VVU IVE can ingest real DWS/municipal SCADA data, detect hydraulic anomalies that correspond to known historical failure events, and produce auditable evidence packages with complete provenance.

Protocol Phases

Phase	Hours	Key Actions	Pass Criterion
A	0-8	Ingest SCADA + failure register + asset metadata	All records imported, no data loss
B	8-24	Run anomaly detection vs historical time series	Anomaly list + match/no-match table
C	24-48	EIS v1.0 evidence assessment per anomaly	Independence score + 11-field provenance
D	48-72	Reporting + DWS engineer review	Detected, lead time, false alerts, missed

Success Criteria

- Events detected at least 48h early: $\geq 50\%$ of confirmed failures
- Median lead time: ≥ 24 hours
- False alerts: < 1 per 10 pipe-months
- Evidence reproducibility: 100% deterministic (same data $\rightarrow$ same result)
- Provenance completeness: 100% of flagged events (all 11 fields)
- Zero Fabrication Rule: no fabricated data, all UNDEFINED preserved

Termination Criteria

The protocol terminates when all 4 phases are complete, OR DWS determines the model does not map correctly, OR 72 hours elapsed. **Either outcome is valid.** The purpose is to measure, not to prove success.

05b · Data Requirements for DWS Validation

Data Needed from DWS

Category	Fields	Purpose
Flow	timestamp, meter ID, value (L/s), quality flags	Anomaly detection
Pressure	timestamp, sensor ID, value (m), quality flags	Anomaly detection
Reservoir level	timestamp, sensor ID, value (m)	Context
Pump/Valve status	timestamp, asset ID, state	Operating context
DMA inflow	timestamp, bulk meter, value (L/s)	Water balance
Min night flow	time series (02:00-04:00)	Baseline computation
Known leak events	asset ID, failure type, location, verification	Validation ground truth
Contextual	rainfall, temperature, power interruptions	False-positive suppression

Success Criteria

- Does system reduce search area? → Yes — narrows to DMA/segment
- Distinguish valid/missing/anomalous/correlated/independent? → Yes — EIS v1.0
- Can engineer inspect why system believes claim? → Yes — provenance drill-down
- Can result be reproduced via pipeline? → Yes — deterministic
- False alert rate → < 1 per 10 pipe-months
- Lead time → ≥ 48 hours before confirmed onset

Data Sharing Agreement

Both parties sign a data sharing agreement covering permitted use, retention period, and de-identification requirements before any real DWS data is provided.

Repository: github.com/divhanimajokweni-ctrl/proofbridge-liner

Branch: feat/vvu-gov-deploy

Domain: vvu-gov.space-z.ai

Classification: SIMULATION — NOT MUNICIPAL OPERATIONAL DATA